

Topic 1: Amazon S3 - Theory and Basic Policy

What is S3?

S3 stands for Simple Storage Service. Think of it as an infinitely large hard drive in the cloud, but instead of folders and files in a traditional filesystem, S3 uses a flat structure of buckets and objects.

Conceptual Comparison

- **Traditional Filesystem:** Hierarchical (/home/user/documents/report.pdf)
- **S3:** Flat structure (my-bucket/documents/report.pdf)

S3 has **no real folders**. "documents/report.pdf" is just a KEY (the full name of the object).

Bucket vs. Object

- **Bucket:** Container for objects. Must be globally unique. Created in a specific Region.
- **Object:** A file + its metadata. Identified by a KEY.
- **Key:** The unique identifier of an object (Bucket name + Key = unique identifier).

Storage Classes

Storage Class	Use Case	Retrieval
S3 Standard	Frequently accessed	Milliseconds
S3 Intelligent-Tiering	Unknown patterns	Milliseconds

S3 Standard-IA	Infrequent access	Milliseconds
S3 One Zone-IA	Infrequent, 1 AZ ok	Milliseconds
S3 Glacier Instant	Archive, instant need	Milliseconds
S3 Glacier Flexible	Archive, hours ok	Minutes/Hours
S3 Glacier Deep Archive	Long-term archive	Hours

Versioning

Without Versioning: Uploading the same key overwrites the original. Deleted files are gone forever.

With Versioning: Keeps all versions. Deleting just adds a "delete marker".

Security Layers

1. IAM Policies
2. Bucket Policies
3. ACLs (Deprecated)
4. Block Public Access
5. Encryption
6. VPC Endpoints

Bucket Policy Basics

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "ExampleStatement",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::my-bucket/*"
    }
  ]
}
```

```
]
}
```

Block Public Access (BPA)

The safety override that prevents accidental public exposure, even if the bucket policy is set to public.